

E-Commerce Sales Analysis – Business Insights

Project Overview

This project analyses an e-commerce sales dataset containing 9,994 transactions using MySQL and Power BI. The objective was to identify sales trends, profitability drivers, customer behavior and regional performance.

Key Findings

1. Sales Performance

- Total Sales: ₹2.30 Million
 - Total Profit: ₹286.40 Thousand
 - Total Orders: 5,009
 - Profit Margin: 12.47%
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2. Category Analysis

Technology generated the highest sales and profit among all product categories.

Sales by Category:

- Technology: ₹0.84M
- Furniture: ₹0.74M
- Office Supplies: ₹0.72M

Profit by Category:

- Technology: ₹0.15M
 - Office Supplies: ₹0.12M
 - Furniture: ₹0.02M
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3. Regional Analysis

The West region contributed the highest revenue.

Sales by Region:

- West: ₹0.73M
- East: ₹0.68M
- Central: ₹0.50M
- South: ₹0.39M

4. Customer Analysis

Top-performing customers generated significantly higher revenue than average customers.

Top Customer:

- Sean Miller: ₹25K+

Customer segmentation shows that the Consumer segment contributes the largest share of profit.

5. State Analysis

California generated the highest sales among all states, followed by New York and Texas.

6. Business Recommendations

1. Increase focus on Technology products due to strong sales and profit performance.
 2. Improve profitability in the Furniture category through pricing and cost optimization.
 3. Expand marketing efforts in high-performing regions such as West and East.
 4. Develop loyalty programs targeting top customers.
 5. Investigate low-performing regions and categories for improvement opportunities.
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Tools Used

- MySQL
- Power BI
- DAX
- CSV Dataset

Dataset Size

- Records: 9,994
- Columns: 25